NEW RADIO COMMUNICATION FOR NEW ERA

NRC7394 Evaluation Kit User Guide

(Channel Manager)

Ultra-low power & Long-range Wi-Fi

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NEWRACOM, Inc.

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NRC7394 Evaluation Kit User Guide (Channel Manager)
Ultra-low power & Long-range Wi-Fi

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1 Overview and prerequisites

The channel manager picks the operating channel automatically when the AP starts, and keeps measuring congestion while the AP runs so it can move to a better channel. It removes the need to pick a channel manually at every installation site and to revisit that choice later.

A brief drop in channel quality does not trigger a switch. The firmware confirms the condition over a period of time before it moves, so it is normal for the channel to stay the same for a while after the feature is enabled.

1.1 Where it can be used

Table 1.1 Conditions for using the channel manager

Item	What to check
Operating mode	Host-mode AP
relay	Not available in relay mode, or on an AP that has a connected relay. A relay stays on its upstream AP's channel, and the current software package does not provide a function that informs a relay of a channel change on the AP.
Country	Not available under country code K2
Channel setting	A fixed channel and automatic selection cannot be used together
Automatic switching	`chmgrd` must run on the host. The Web GUI and an enabled startup setting launch it automatically

1.2 Effect on service

Data transmission is briefly restricted during a measurement, so throughput can drop. `monitor` mode performs the same measurements and has the same effect. When the channel changes, the associated STAs receive the switch announcement and follow the AP to the new channel.

Start with the default profile `normal` and observe the actual link behavior. With a short beacon interval, some beacon transmissions can be skipped during a measurement.

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2 Quick start

2.1 Turning on automatic channel selection from the Web GUI

1. Open the Web GUI over the wired management address.
2. Check the operating mode and the country under **Settings**.
3. Set **Channel** under **Network** to `auto`.
4. Check the channel width that is shown. Unless you have a specific requirement, leave the measurement time and the exclusion list as they are.
5. Save the settings and follow the application notice on the screen. If the wireless function is stopped, start it from the dashboard.
6. Confirm the channel actually in use from the dashboard.

Settings saved from the GUI survive a wireless restart and a device reboot. Applying settings to a running AP can interrupt the link, so read the application notice first.

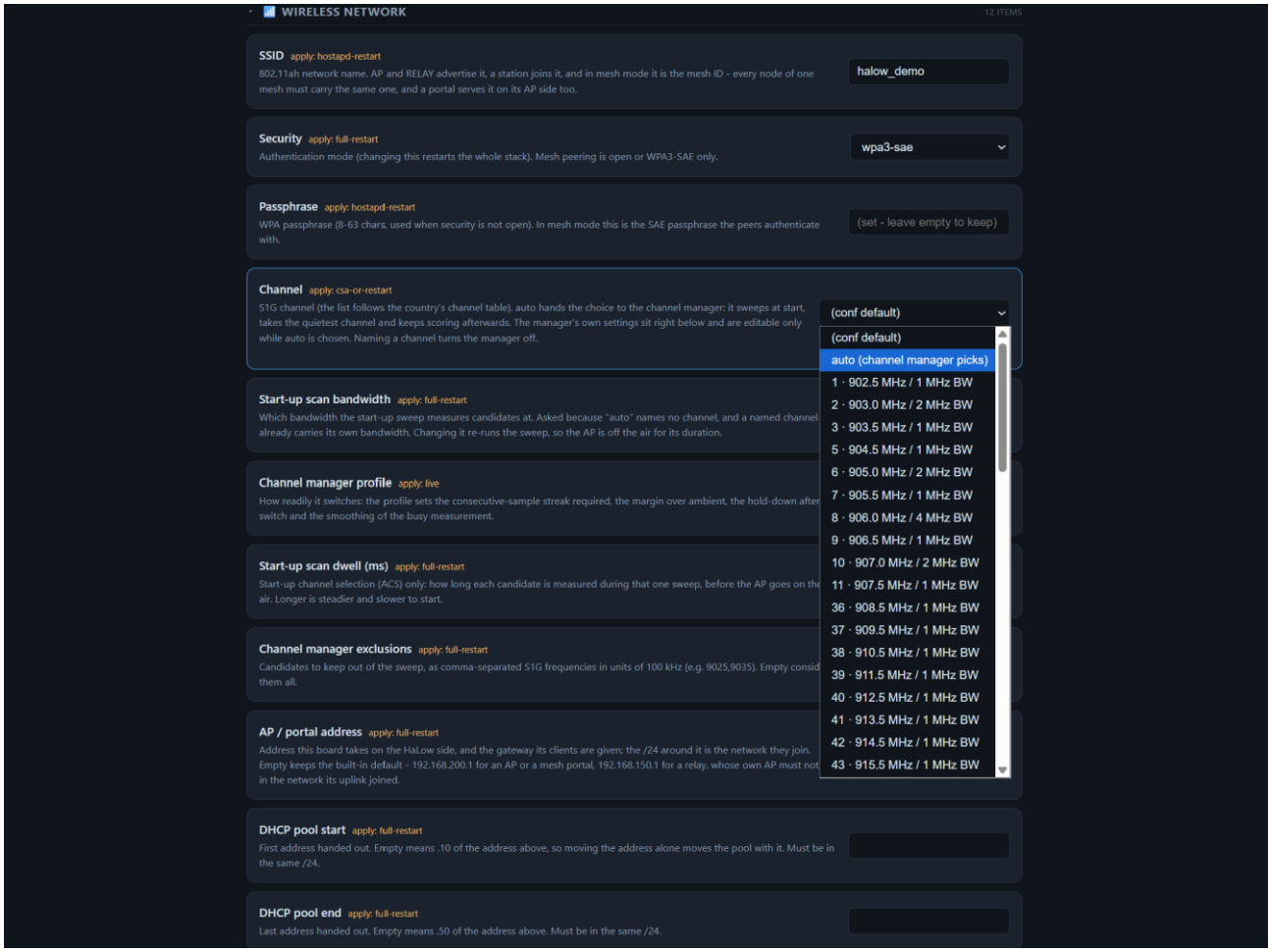


Figure 2.1 Selecting automatic channel management

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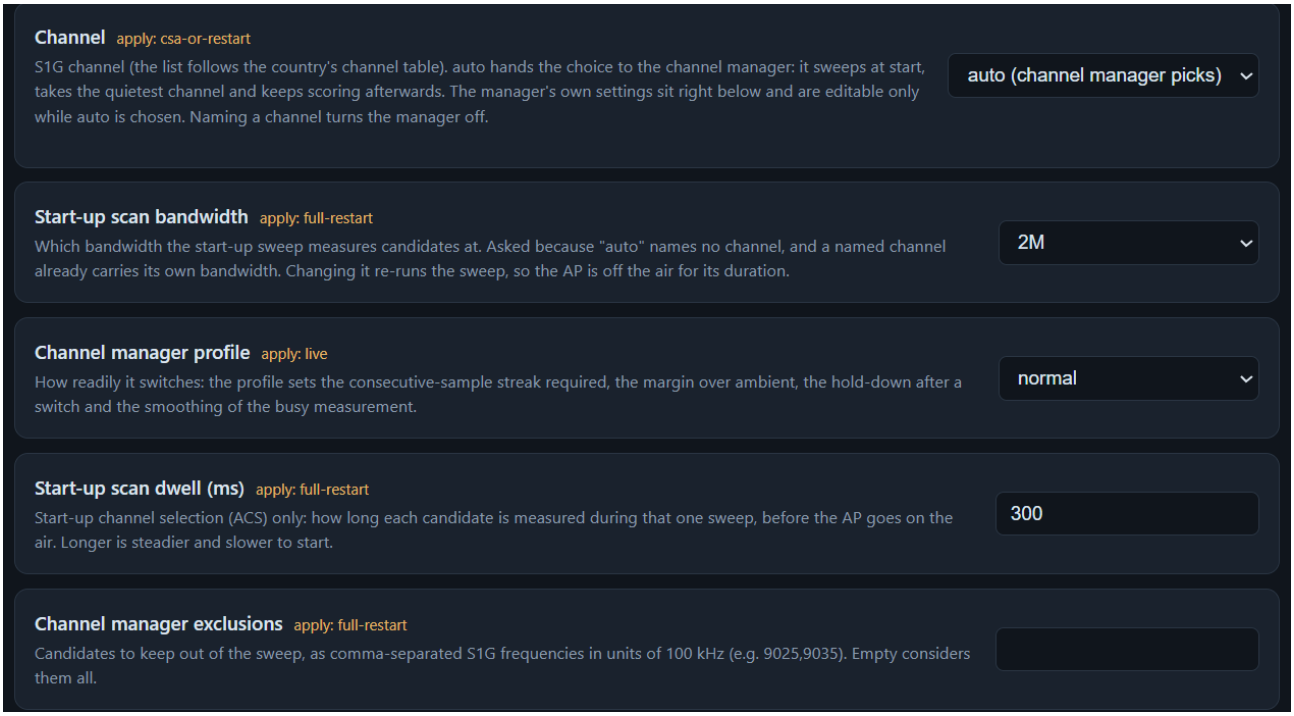


Figure 2.2 Channel manager settings with automatic selection enabled

2.2 Watching without changing the channel

To look at the measurement results first on an AP that is already running, use the CLI.

```
cd ~/nrc_pkg/script
./cli_app set chmgr monitor profile=1 cc=<country> bw=<width>
./cli_app show chmgr
```

With `mode=monitor` the channel is never switched automatically. Read the output again after some time and compare the candidate scores. Section 3.2 describes how to move from `monitor` to `auto`. Give cc= and bw= the country code and the channel width (1, 2 or 4 MHz) the AP is running with. Without bw= the candidate list is built for 1 MHz, so the current channel shown does not match a 2 MHz or 4 MHz AP.

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3 Configuration

3.1 Operating mode and profile

Table 3.1 Operating modes

Mode	Behavior	Use it for
`off`	Measurement stopped, current channel kept	Stopping the channel manager
`monitor`	Measures only, never switches	Observing the site
`auto`	Measures and switches when the conditions are met	Automatic channel management

The profile decides how cautiously a switch is made.

Table 3.2 Profiles

Value	Profile	Choose it when
0	`conservative`	You want fewer switches and less reaction to short changes
1	`normal`	General use. Default profile
2	`aggressive`	You want a faster reaction to changing conditions

No profile guarantees an immediate switch. Depending on the measurements and the extra confirmation step, a switch can be canceled or delayed.

3.2 Turning on automatic switching on a running AP

When the feature is enabled from the CLI, `chmgrd` must also be running on the host. The example below assumes the AP interface is `wlan0`.

```
cd ~/nrc_pkg/script
if ! pgrep -x chmgrd >/dev/null; then
    if sudo -v; then
        sudo -n ~/nrc_pkg/script/chmgrd -i wlan0 &
    fi
fi
./cli_app set chmgr auto profile=1
./cli_app show chmgr
```

Confirm `mode=auto` and `state=running`. A CLI change is not written to the startup settings, so use the Web GUI if the setting has to survive a restart.

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3.3 Stopping

```
./cli_app set chmgr off
```

Measurement stops and the AP keeps its current channel. show chmgr then reports state=disabled and mode=off. The pick and s1g values on its first line remain the start-up selection rather than the channel in use, and the candidate list returns to the start-up scan values. This command alone does not change the startup settings. To keep using a fixed channel, select that channel in the Web GUI and save it.

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4 Checking operation

4.1 Current channel and measurement state

```
./cli_app show chmgr
```

Check these items first.

Table 4.1 Main items in the show chmgr output

Item	How to read it
`state`	`running` once the AP has started. `armed` means waiting for the start, `disabled` means the channel manager is off
`mode`	`auto` for automatic switching, `monitor` for observation only
`cur`	Current operating frequency. `9050` means 905.0 MHz
Candidate marked with `*`	The current operating channel and its score
Number in parentheses in the candidate list	Score derived from CCA occupancy. A lower value means a less congested candidate
`eval`	Number of completed evaluations. Check that it keeps increasing over time

`pick`, `s1g` and `cca` on the first line are the selection made at startup, not the current values after a switch. **Read the current channel from `cur` and from the candidate marked with `*`.**

Seeing `dcs=1(monitor)` while `mode=auto` is normal. `mode` is the operating mode you selected and `dcs` is the current processing state. Even in automatic mode the engine spends most of its time observing the channel.

```
pi@raspberrypi:~/nrc_pkg/script $ sudo ./cli_app show chmgr
state=running mode=auto ncand=25 bw=2 pick=2 s1g=9030 cca=3.1 wdt=0
dcs=1(monitor) cur=9030 streak=0/10 eval=14 sw=0 veto=0 req=0 last=9040
*903.0(0.4)904.0(3.7)927.0(4.6)905.0(4.8)925.0(5.2)918.0(6.4)913.0(6.7)906.0(6.9)911.0(7.9)915.0(8.1)912.0(9.3)924.0(9.3)921.0(9.7)
919.0(11.3)914.0(12.1)920.0(12.6)922.0(13.1)909.0(14.7)908.0(15.2)916.0(15.2)907.0(17.0)926.0(30.0)910.0(32.4)917.0(32.6)923.0(44.2
)
```

Figure 4.1 Example channel manager CLI status

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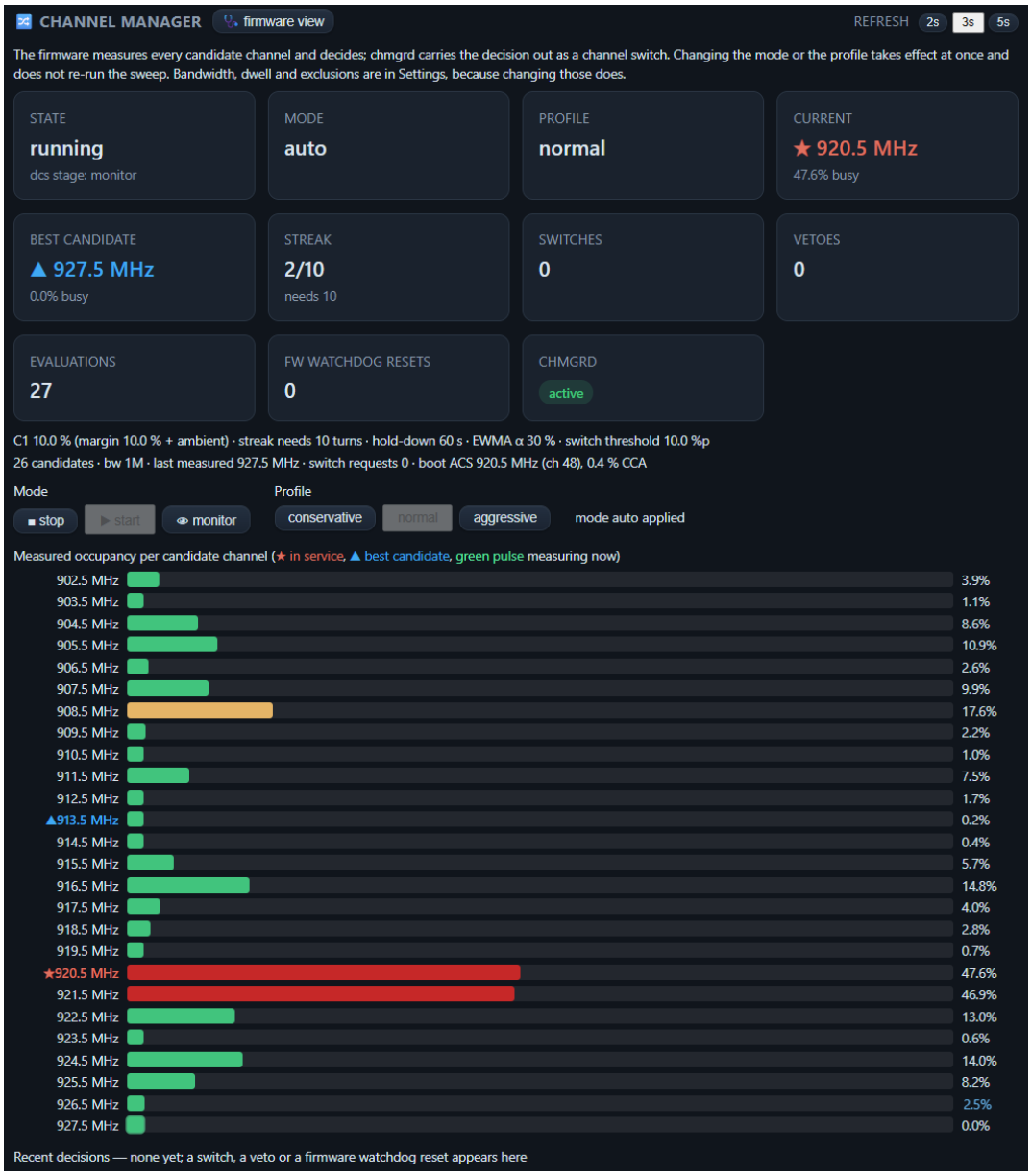


Figure 4.2 Channel manager dashboard with candidate occupancy results

4.2 Confirming a completed switch

A lower score on another candidate does not cause an immediate switch. The firmware confirms that the advantage holds for a period of time, which can take several minutes depending on the profile and the environment.

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To confirm that a switch happened, compare `cur` before and after. Check that it changed to the requested target frequency and that `req` returned to 0, then verify traffic on the associated STAs.

`sw` counts switch **requests**. `chan_switch OK` in the log only means the command was accepted, so do not judge a completed switch from these two values alone.

Figure 4.3 shows the dashboard after a completed switch: SWITCHES counts up, the stage shows holddown, and the switch appears under Recent decisions.

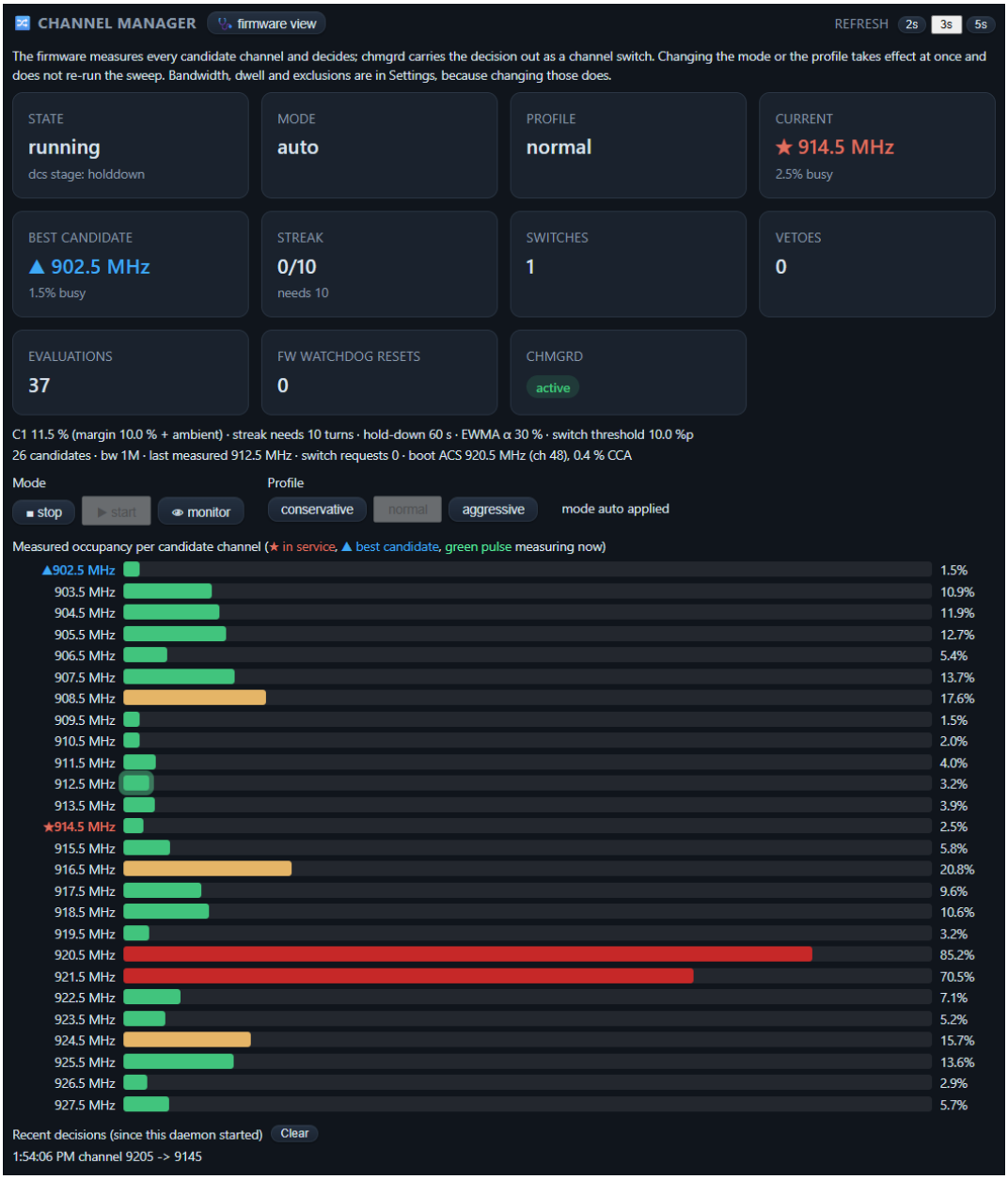


Figure 4.3 Channel manager dashboard after a completed switch

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5 Troubleshooting

5.1 The feature cannot be turned on

Check the operating mode, the country code, whether a relay is connected, and any user RAW setting. Also check that a fixed channel and automatic selection were not requested together. If `fail(-5)` is shown, check whether a user RAW is active.

5.2 Another channel looks better but no switch happens

Read the `show chmgr` output and check the following in order.

Table 5.1 show chmgr indications and actions

Shown	Action
`mode=monitor`	Observation-only mode. If automatic switching is needed, confirm that `chmgrd` runs and change the mode to `auto`.
`n` below `N` in `streak=n/N`	The required number of qualifying evaluations has not been reached. Check again later.
`dcs=2(verify)`	The extra confirmation before a switch is in progress. Wait for it to finish.
`dcs=4(holddown)`	The waiting period after the previous switch.
`veto` increasing	The confirmation canceled the switch. The target channel did not keep its advantage.
`req` not 0	A switch request is being processed. If the request remains pending, check the host state.

5.3 A switch request never completes

Check the daemon and the AP with the following commands.

```
pgrep -x chmgrd
sudo tail /tmp/chmgrd.log
sudo hostapd_cli -i wlan0 status
```

- No output from the first command means the daemon is not running. Start it as described in section 3.2.
- If `chan_switch FAILED` appears, read the failure reason logged with it. Check that the AP is running, that the interface name is correct, and that the daemon has root privileges.
- A value such as `2437 MHz` in the log can be the frequency mapped for control purposes. Read the actual HaLow operating frequency from `cur` in `show chmgr`.

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If the problem persists, send the `show chmgr` output and `/tmp/chmgrd.log` to technical support.

5.4 Scores stop updating, or throughput drops

Heavy uplink traffic can cause some measurements to be discarded, delaying score updates. The measurement itself also restricts data transmission briefly. If needed, compare against the same setup with the feature set to `off` to see the effect on service.

Detailed logs for technical support are enabled as follows. Turn them off again afterwards.

```
./cli_app set chmgr log 1
./cli_app set chmgr log 0
```

These two commands are not meant to be run back to back. Enable the log with the first command, reproduce the symptom, and turn it off with the second command once the data is collected.

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6 Advanced settings and CLI reference

6.1 Candidate channel settings

```
set chmgr {off|auto|monitor} [profile=0|1|2]
        [cc=CC] [bw=1|2|4] [exclude=<s1g,..>]
set chmgr log {0|1}
```

Run the syntax above after `./cli_app`.

Table 6.1 set chmgr arguments

Argument	Meaning
`profile=`	0 conservative, 1 normal, 2 aggressive
`cc=`	Country code applied to the candidate list. Match it to the installation region and the device setting
`bw=`	Candidate channel width in MHz. Only 1, 2 and 4 are accepted. A new scan defaults to 1 MHz if omitted; a mode/profile change or warm re-enable without candidate overrides retains the existing candidate set
`exclude=`	S1G frequencies to exclude, in units of 0.1 MHz, separated by commas. Example: `9025,9035`
`log`	Per-measurement log. 0 off, 1 on. Off by default

When candidates are set from the CLI, use values that match the current country and channel width settings.

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6.2 Start-up settings from the script

When `start.py` is used instead of the Web GUI, set the feature in its header. The shipped default is channel_manager = 0 (manual). Set prefer_bw to the channel width in use.

```
channel_manager           = 1   # 0: manual (off), 1: auto, 2: monitor
channel_manager_profile    = 1   # 0: conservative, 1: normal, 2: aggressive
channel_manager_prefer_bw = 1   # candidate bw: 1: 1M, 2: 2M, 4: 4M
channel_manager_dwell     = 300 # boot-scan dwell per channel (ms, 100..1000)
channel_manager_exclude    = ''  # S1G freq x10 to avoid, e.g. '9025,9035'
```

With the channel manager enabled in the startup settings, `start.py` launches `chmgrd` automatically. If a fixed channel argument is given on that run, the fixed channel wins and the channel manager stays disabled.

```
################################################################################
# Channel Manager Conf. (AP Only) - boot ACS + runtime DCS
#  0 (manual) : use channel= in hostapd conf as-is (engine off)
#  1 (auto)   : FW picks the start channel (boot ACS) and keeps switching
#               it by channel quality in service (DCS)
#  2 (monitor): keep the manual channel; engine measures and logs only
channel_manager           = 1
channel_manager_profile   = 1   # 0: conservative, 1: normal, 2: aggressive
channel_manager_prefer_bw = 2   # candidate bw (explicit): 1: 1M, 2: 2M, 4: 4M
channel_manager_dwell     = 300 # boot-ACS scan dwell per channel (ms, 100..1000; 300 recommended)
channel_manager_exclude   = ''  # comma-separated S1G freq x10 to avoid (e.g. '9025,9035')
################################################################################
```

Figure 6.1 Channel manager startup settings in start.py

6.3 Daemon options

The daemon normally starts automatically, so there is usually no need to run it manually. A manual run requires root privileges.

Table 6.2 chmgrd options

Option	Default	Meaning
`-i <iface>`	`wlan0`	AP interface
`-l <file>`	`/tmp/chmgrd.log`	Log file

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7 Revision history

Version	Date	Description
Ver 1.0	9/28/2026	Initial release

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